Master table

ci-tec | README | 13.10.2026 | Master schema v7 | Time policy v4

Use **Bio_O_Ton_Master.csv**, one row per recording. No separate ci-tec table is required. This is the latest recorded inventory, not a guarantee that processing is finished. Original metadata source: dawn-chorus-soundscape.csv.

Column	Meaning
dawn_chorus_id	Recording ID.
lon, lat	WGS84 coordinates in decimal degrees.
grid_100m_id, grid_10m_id	Assigned grid cells. A grid ID does not imply that LRT formation data are available.
datetime_local	German local date and clock time in Europe/Berlin, stored without a timezone suffix. Its origin is recorded separately in timestamp_source.
datetime_utc	UTC instant derived from the selected German recording clock. A misleading +00:00 datetime label does not shift a German wall clock.
timestamp_source	Exact values: localtimes for a supplied local recording clock; datetime for a source fallback. The source field is provenance, not a QC failure.
time_has_issues	False for a successfully validated time from either source. True for an unresolved, missing or conflicting time. Source datetime alone is not an issue.
duration_seconds	Checked audio duration in seconds; null if unavailable.
inside_lrt_polygon	Recording point lies within an included LRT polygon.
grid_100m_has_majority_formation, grid_10m_has_majority_formation	Majority formation is available for this cell; different from a direct point/LRT intersection.

Grid embeddings, the LRT GeoPackage and grid reference data are separate products. Recording/bioacoustic flags do not certify their completeness.

Recording time: source and German clock

The Dawn Chorus field **localtimes** has priority. Only if missing is **datetime** used. In the audited zero-offset (+00:00/Z) German fallback class, datetime stores the German local *clock*, not a UTC instant. Localize the clock to **Europe/Berlin** without first shifting the hour and then derive real UTC. Summer/winter rules and DST gaps/folds must be checked.

Examples without localtimes: 2025-01-15T06:00:00Z becomes 2025-01-15 06:00:00 (UTC 05:00Z); 2025-07-15T06:00:00Z becomes 2025-07-15 06:00:00 (UTC 04:00Z).

When localtimes is supplied, its local clock retains priority and is interpreted as German time. These recordings are unchanged by the fallback rule. A nonempty but invalid or unresolved localtimes value is flagged; it is not silently replaced with datetime. Source conflicts remain visible in the Step-1 time log.

Revised audit on 13.10.2026: the old UTC conversion was wrong for 3,288 datetime fallback records: 3,264 had a +2 h and 24 a +1 h clock shift. All 3,288 corrected clocks match the source fields; 14 website examples independently corroborate the rule (not every row was checked at the website). Four local dates and one UTC date changed. The 47 separate localtimes/UTC conflicts retain issue flags.

Clean-data filters from the 9 October PDF

Apply the filters needed by your analysis:

```
time_has_issues=False
sound_exists=True
duration_seconds >= 40
sound_has_issues=False
weather_point_exists=True
weather_point_has_issues=False
sentinel_exists=True
sentinel_has_issues=False
```

The 40-second threshold is an analysis choice and can be replaced. Domain flags

do not automatically invalidate the entire recording. For training in cells with

a majority formation add grid_100m_has_majority_formation=True; for direct

point-in-LRT training add inside_lrt_polygon=True. Grid IDs alone certify neither.

Recording-time confidence and weather use

Use weather data only for recordings with time_has_issues=False. This is a validation flag, **not** a source-provenance restriction; a validated datetime fallback is allowed. Genuine time conflicts remain flagged. Four weather windows require a date-based recheck.

This timestamp validation supplements weather availability and quality checks. In schema v7, combined readiness flags include it. It does not automatically exclude the recording from audio-only work, and the audio-only bioacoustic readiness flag remains independent of time confidence. Recordings and stored files remain in the shared master.

Availability and issues

_exists: True means the corresponding files were recorded as present on LSDF. These flags describe the recorded inventory, not a live file check.

_has_issues: True means a problem or missing required data/check results. It can be True whether _exists is True or False.

Empty means no check result is available.

_issue_codes gives the reason. Multiple codes are separated by a pipe; details may follow a colon. **_has_issues is not a severity level or a universal exclusion rule.** Rules below concern the affected file/domain, not automatically the entire recording. Unknown codes or a True issue flag without codes require review.

Columns	Data
sound_exists, sound_has_issues	Audio file and audio checks.
sentinel_exists, sentinel_has_issues	Sentinel-2 files and technical checks.
weather_point_exists, weather_point_has_issues	Weather file for the recording and its checks.

Sound examples:

• True / False: file found, no issues reported.

• True / True: file found, but a technical property or decoding check failed.

• False / True: file missing.

Sound codes and suggested use

sound_issue_codes	Meaning / consequence
duration_not_60s	Outside 59-61 seconds. **Information**; decide in application-specific preprocessing. Not a general exclusion.
missing_file	File absent: unavailable.
empty_file, no_audio_stream_found, no_audio_frames_decoded	No usable audio in the checked file: exclude that file pending repair.
pyav_decode_failed, invalid_or_missing_sample_rate, invalid_or_missing_channel_count, invalid_duration, duration_unavailable	Technical check failed or a property could not be established: hold for review. Does not prove the original source is permanently corrupt.
multiple_audio_streams	Select the intended stream before use.
filename_does_not_match_<id>_audio_pattern	File-to-ID association needs clarification.
source_stat_failed, copy_failed, WinError	Access/copy problem; investigate connection and download before concluding source corruption.

PDF audio filter: sound_exists=True, duration_seconds >= 40, sound_has_issues=False. A pure duration_not_60s note does not set the issue flag. Shorter or longer files can be usable; change the duration threshold for your analysis. Technical codes still require review.

Point weather codes and suggested use

Expected window: ten preceding complete German calendar days plus the recording day, hourly. Normally 264 rows; across spring DST 263, across autumn DST 265. The repeated local autumn hour is valid; a timezone-free timestamp alone cannot distinguish its two UTC instants.

weather_point_issue_codes	Meaning / consequence
missing_file	No weather file. Cause is not established by this code.

weather_point_issue_codes	Meaning / consequence
missing_value	At least one required value is missing. **Information only:** this code alone does not set weather_point_has_issues. Check variables/hours needed by the application; do not fill values blindly.
recording_date_changed_recheck_required	Recording date changed while previous results were retained. Temporal association is unconfirmed: hold until rechecked.
unexpected_time_interval	Full timestamp sequence and multiplicities differ from the expected window. May mean missing, extra or shifted hours. Correct DST transitions are accepted by the current inventory. **Check/correct before use.**
unexpected_time_window, unexpected_row_count	Wrong boundaries or row count: check/correct before use.
duplicate_datetime	More repeats than allowed by the expected window, not the legitimate repeated autumn hour. Check/correct before use.
missing_required_column, missing_datetime_column, unparseable_datetime, invalid_metadata_datetime	Structure or time basis is unreliable: hold until clarified.
empty_file, read_error	Empty or unreadable; an exception suffix may follow read_error. File currently unusable.
implausible_value	Temperature/humidity range check failed. Investigate values and source before deciding usability.

PDF weather filter: time_has_issues=False, weather_point_exists=True and weather_point_has_issues=False. This permits missing_value as an informational code; verify the variables/hours your analysis needs. A stricter application can additionally require no codes. Additional time/structure errors remain blocking.

Sentinel

Code / field	Meaning / consequence
Sentinel: missing_file, score_without_tif	Image missing; a score does not replace it.
Sentinel: too_cloudy_sentinel_not_available	Quality score is exactly 0: too cloudy, Sentinel data unavailable for use. The numeric score stays 0; sentinel_has_issues=True. sentinel_exists still reports physical file presence.
Sentinel: quality_score_missing	Image may exist, but its quality assessment is missing.
Sentinel: empty_file, raster_read_failed	Image empty or unreadable: exclude pending repair.
inventory_not_run, qc_not_run	Check results unavailable; not equivalent to passing QC.

Weather and Sentinel do not automatically exclude a recording from audio/LRT-only work. Photos are not a filter criterion for the stated ci-tec priorities.

Processing completion is separate from quality

is_final does not currently exist. ready_for_general_analysis is not a ci-tec release flag: it requires trusted time provenance plus unflagged sound, weather and Sentinel together. It does not enforce an analysis-specific audio duration threshold; apply duration_seconds >= 40 separately for the PDF clean filter.

Completion requires finished processing and checks for the required products, a specified input/code version and no outstanding repairs. Missing data and suitability remain separate. A permanent release applies only to a frozen version; later versions can reveal new errors. The shared master retains all rows.